Spaceability of Smooth Disk-Algebra Functions with Boundary Pringsheim Singularities

Math discovery smoke-test packet

June 12, 2026

1 Source

The source paper is

Andreas Debrouwere, *Some results about spaceability in function spaces*, arXiv:2606.13195.

Remark 3.15 points to the following question from Bernal-Gonzalez et al.:

$\mathbb{N}$. $\square$

Remark 3.15. Corollary 3.14 sheds some light on the question on [6, p. 607], which asked whether the smaller set

$$\{f \in A^\infty(\mathbb{D}) \mid \widetilde{f_{|\mathbb{T}}} \text{ is Pringsheim singular at every point of } \mathbb{R}\}$$

is spaceable in $A^\infty(\mathbb{D})$.

Here $A^\infty(\mathbb{D})$ denotes the smooth disk algebra. A boundary point is Pringsheim singular for the boundary restriction when the holomorphic function has no analytic continuation through that point. Thus it is enough to construct a closed infinite-dimensional subspace of $A^\infty(\mathbb{D})$ whose nonzero elements all have $\mathbb{T}$ as a natural boundary.

2 Claim

Theorem 1. *The set*

$$\{f \in A^\infty(\mathbb{D}) : f|_{\mathbb{T}} \textit{ is Pringsheim singular at every boundary point}\}$$

is spaceable in $A^\infty(\mathbb{D})$.

3 Proof

We use the classical Hadamard gap theorem in the following form.

Lemma 1 (Hadamard gap theorem)**.** *Let*

$$F(z) = \sum_{\ell=1}^{\infty} a_\ell z^{m_\ell}$$

have radius of convergence 1*, where* $m_{\ell+1}/m_\ell \geq q > 1$*. Then the unit circle is a natural boundary for* F*.*

Let the master lacunary sequence be $N_k = 2^k$, $k \geq 1$. Partition $\mathbb{N}$ into infinitely many infinite sets

$$A_m = \{2^m(2j+1) : j = 0, 1, 2, \ldots\}, \qquad m = 0, 1, 2, \ldots.$$

For $m \geq 0$, define

$$n_{m,j} = 2^{2^m(2j+1)}$$

and

$$g_m(z) = \sum_{j=0}^{\infty} \exp(-\sqrt{n_{m,j}})\, z^{n_{m,j}}.$$

The sets $\{n_{m,j} : j \geq 0\}$ are pairwise disjoint and all lie in the master support $\{2^k : k \geq 1\}$. Moreover,

$$\sum_j n_{m,j}^r \exp(-\sqrt{n_{m,j}}) < \infty$$

for every $r \geq 0$. Hence each derivative series of g_m converges uniformly on $\overline{\mathbb{D}}$, so $g_m \in A^\infty(\mathbb{D})$.

Let

$$X = \overline{\operatorname{span}\{g_m : m = 0, 1, 2, \ldots\}}^{A^\infty(\mathbb{D})}.$$

This is a closed infinite-dimensional subspace, since the g_m's have pairwise disjoint nonzero Taylor supports and are therefore linearly independent.

We claim that every nonzero $f \in X$ has $\mathbb{T}$ as a natural boundary. Write

$$f(z) = \sum_{n=0}^{\infty} c_n z^n.$$

The coefficient functionals $f \mapsto c_n = f^{(n)}(0)/n!$ are continuous on $A^\infty(\mathbb{D})$. Since finite linear combinations of the g_m's have no coefficients outside the master set, every $f \in X$ also satisfies

$$c_n = 0 \qquad (n \notin \{2^k : k \geq 1\}).$$

Likewise, for fixed m, finite linear combinations satisfy

$$\frac{c_{n_{m,j}}}{\exp(-\sqrt{n_{m,j}})} = \frac{c_{n_{m,j'}}}{\exp(-\sqrt{n_{m,j'}})} \qquad (j, j' \geq 0),$$

and this equality is preserved under closure. Therefore each $f \in X$ has scalars λ_m such that

$$c_{n_{m,j}} = \lambda_m \exp(-\sqrt{n_{m,j}}) \qquad (j \geq 0).$$

If $f \neq 0$, then $\lambda_m \neq 0$ for at least one m; otherwise all Taylor coefficients of f would be zero.

For such an m,

$$\lim_{j\to\infty} |c_{n_{m,j}}|^{1/n_{m,j}} = \lim_{j\to\infty} |\lambda_m|^{1/n_{m,j}} \exp(-1/\sqrt{n_{m,j}}) = 1.$$

Since $f \in A^\infty(\mathbb{D})$, its radius of convergence is at least 1. The preceding limit shows that the radius is exactly 1. The nonzero Taylor support of f is a subsequence of the master support $\{2^k\}$; therefore, when listed increasingly, its successive exponents have ratio at least 2. Hadamard's gap theorem applies and gives $\mathbb{T}$ as a natural boundary for f. Thus $f|_{\mathbb{T}}$ is Pringsheim singular at every boundary point.

Therefore $X \setminus \{0\}$ is contained in the set in the theorem, and the set is spaceable. $\square$

4 Verification Notes

The script `code/check_lacunary_construction.py` checks finite samples of the partition and coefficient profile:

```
conda run --no-capture-output -n sandbox python code/check_lacunary_construction.py
```

The run confirmed disjoint blocks, a master consecutive ratio at least 2, and the expected root behavior of $\exp(-\sqrt{n})$. These checks are only sanity checks; the proof above is analytic and uses Hadamard's theorem.

5 Limitations and Review Focus

The main external ingredient is the classical Hadamard gap theorem. A human reviewer should verify that the terminology "Pringsheim singular at every point" in Bernal-Gonzalez et al. agrees with the natural-boundary conclusion used here. The construction gives the stronger conclusion that no nonzero element of X analytically continues through any boundary point.